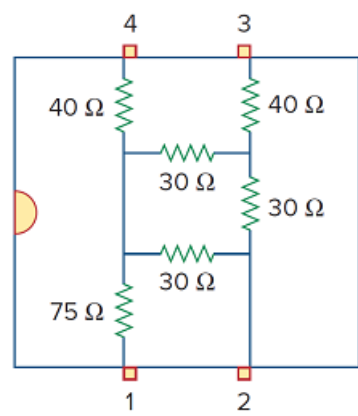


Problem

The pin diagram of a resistance array is shown in Fig. 2.141. Find the equivalent resistance between the following:

- (a) 1 and 2
- (b) 1 and 3
- (c) 1 and 4

Figure 2.141



Step-by-step solution

Step 1 of 17

Consider the following resistance array circuit diagram:

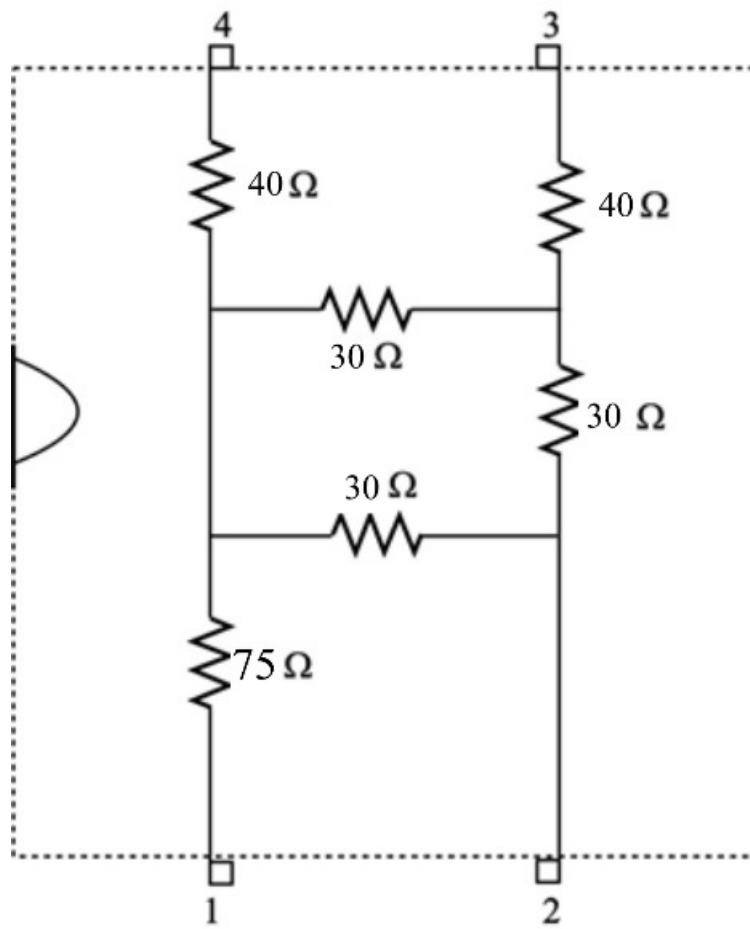


Figure 1

(a)

Consider the following circuit diagram to calculate equivalent resistance between 1 and 2:

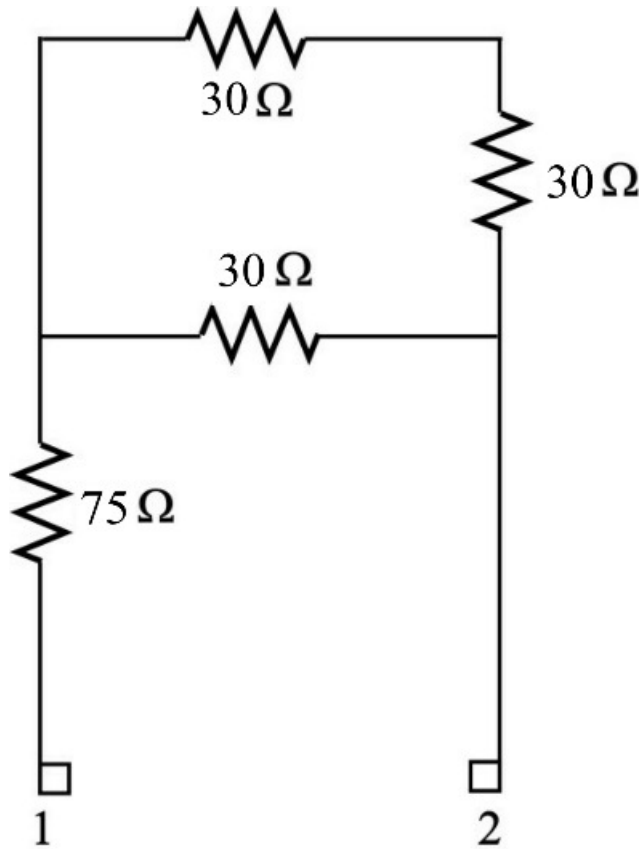


Figure 2

Step 3 of 17

From Figure 2, the resistances $30\ \Omega$ and $30\ \Omega$ are in series.

Calculate the series equivalent resistance is as follows.

$$\begin{aligned}\text{Equivalent series resistance} &= 30\ \Omega + 30\ \Omega \\ &= 60\ \Omega\end{aligned}$$

Step 4 of 17

The modified diagram of Figure 2 is shown in Figure 3:

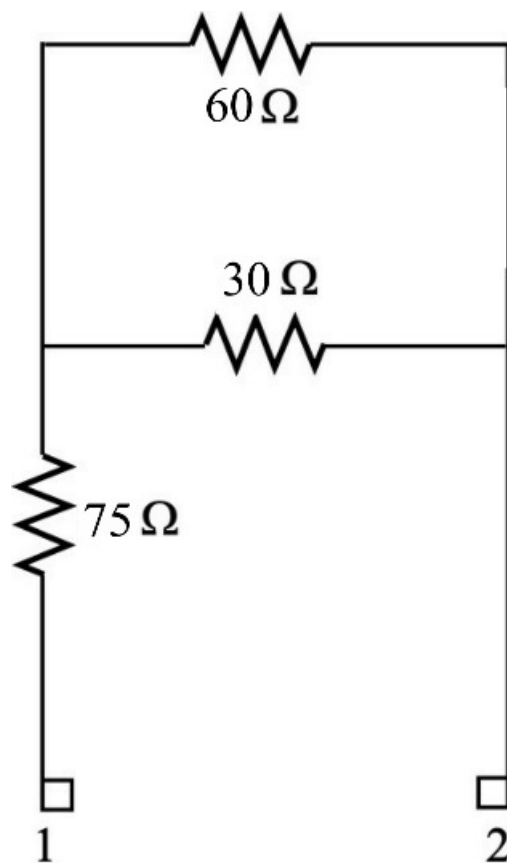


Figure 3

Step 5 of 17

From Figure 3, the resistances $60\ \Omega$ and $30\ \Omega$ are in parallel.

Calculate the parallel equivalent resistance is as follows.

$$\begin{aligned}(60\ \Omega \parallel 30\ \Omega) &= \frac{(60\ \Omega)(30\ \Omega)}{60\ \Omega + 30\ \Omega} \\ &= \frac{1800\ \Omega}{90\ \Omega} \\ &= 20\ \Omega\end{aligned}$$

Step 6 of 17

The modified diagram of Figure 3 is shown in Figure 4:

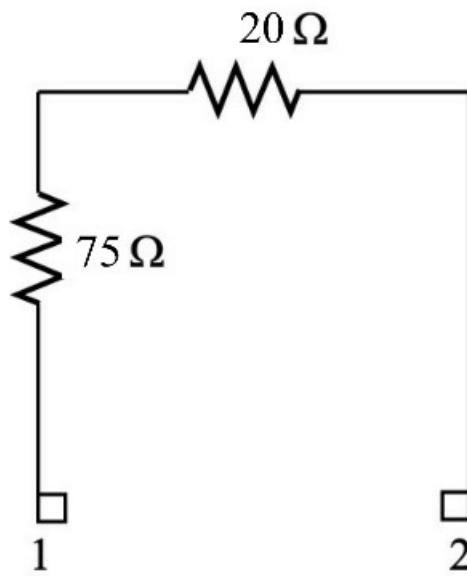


Figure 4

Step 7 of 17

From Figure 4,

Calculate the equivalent resistance between 1 and 2 are as follows.

$$\begin{aligned}\text{Equivalent resistance} &= 75\ \Omega + 20\ \Omega \\ &= 95\ \Omega\end{aligned}$$

Therefore, the equivalent resistance between 1 and 2 is $\boxed{95\ \Omega}$.

Step 8 of 17

(b)

Consider the following circuit diagram to calculate equivalent resistance between 1 and 3:

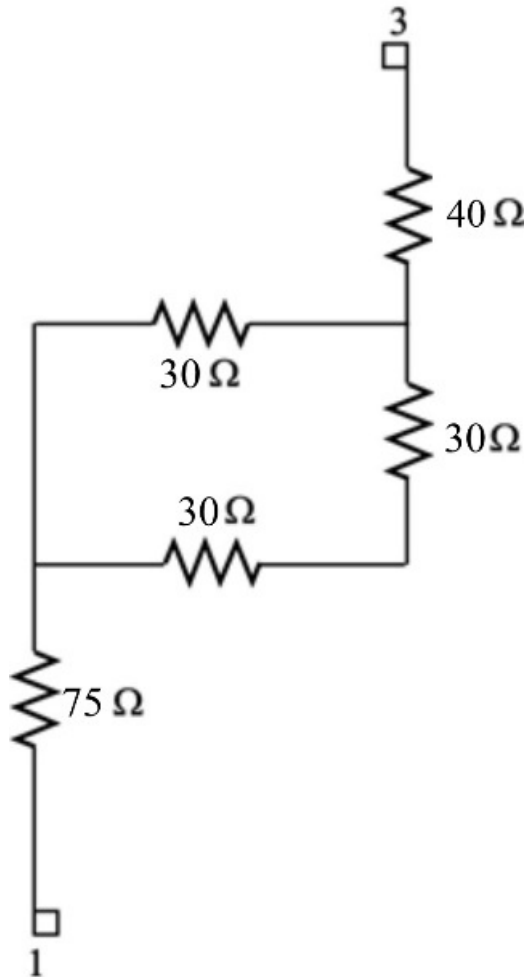


Figure 5

Step 9 of 17

From Figure 5, the resistances $30\ \Omega$ and $30\ \Omega$ are in series.

Calculate the series equivalent resistance is as follows.

$$\begin{aligned}\text{Equivalent series resistance} &= 30\ \Omega + 30\ \Omega \\ &= 60\ \Omega\end{aligned}$$

Step 10 of 17

The modified diagram of Figure 5 is shown in Figure 6:

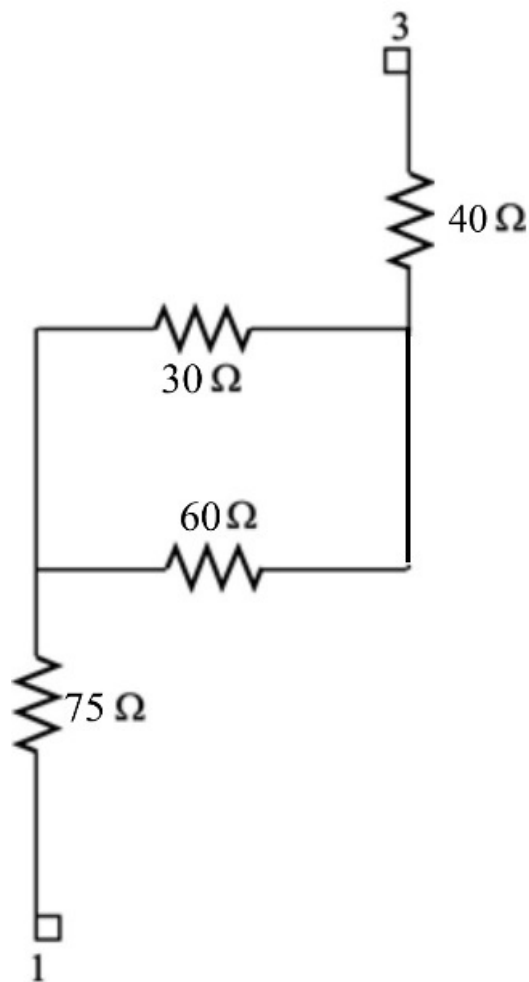


Figure 6

Step 11 of 17

From Figure 6, the resistances 30Ω and 60Ω are in parallel.

Calculate the parallel equivalent resistance as follows.

$$\begin{aligned}(60\Omega \parallel 30\Omega) &= \frac{(60\Omega)(30\Omega)}{60\Omega + 30\Omega} \\ &= \frac{1800\Omega}{90\Omega} \\ &= 20\Omega\end{aligned}$$

Step 12 of 17

The modified diagram of Figure 6 is shown in Figure 7:

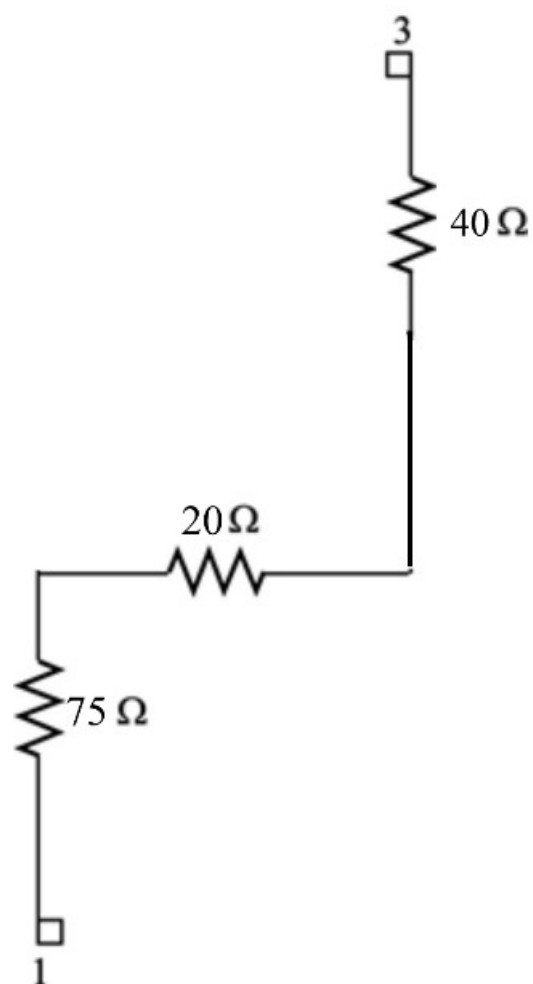


Figure 7

From Figure 7,

Calculate the equivalent resistance between 1 and 3 are as follows.

$$\begin{aligned}\text{Equivalent resistance} &= 75\Omega + 20\Omega + 40\Omega \\ &= 135\Omega\end{aligned}$$

Therefore, the equivalent resistance between 1 and 3 is 135Ω .

(c)

Consider the following circuit diagram to calculate equivalent resistance between 1 and 4:

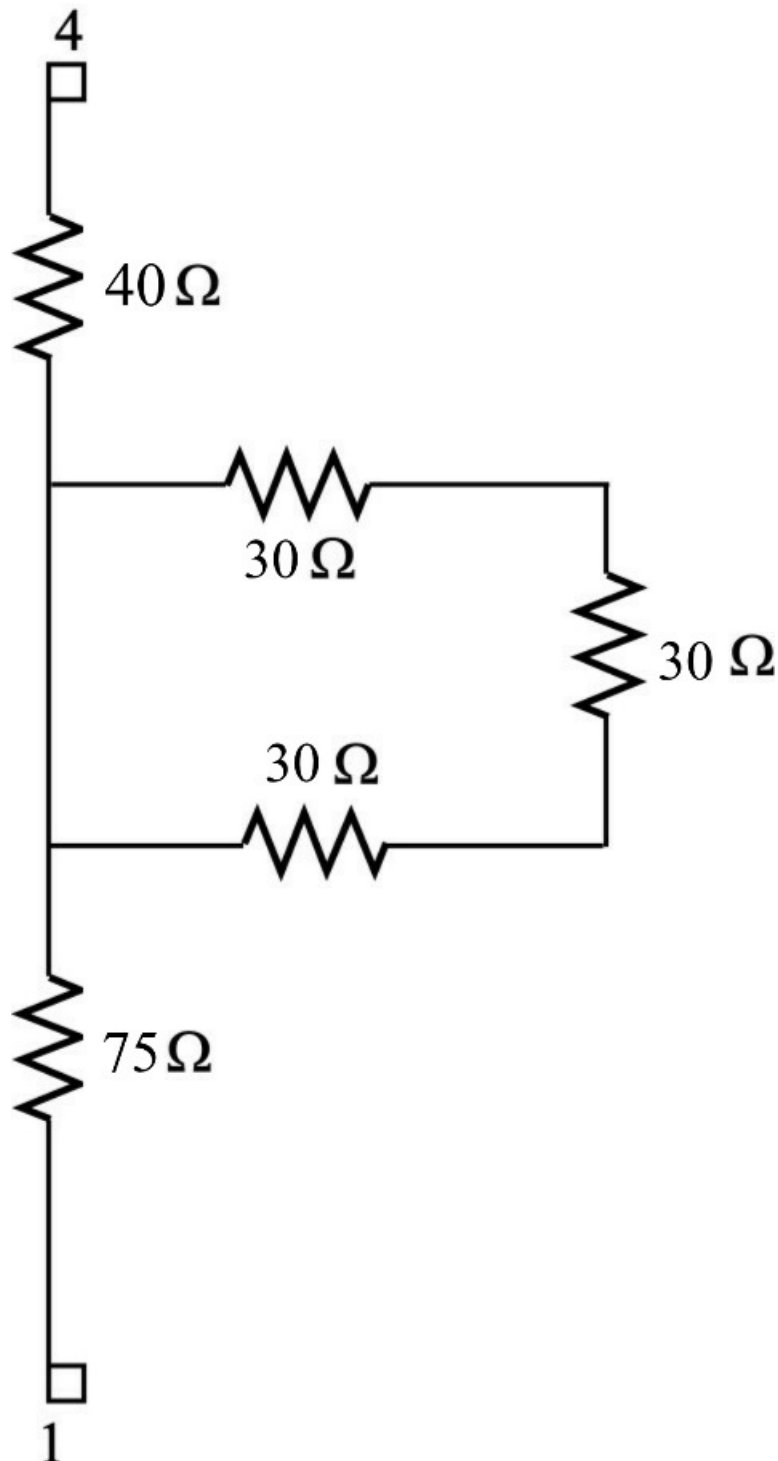


Figure 8

From Figure 8, the resistances 30Ω , 30Ω and 30Ω are in series.

Calculate the series equivalent resistance is as follows.

$$\begin{aligned}\text{Equivalent series resistance} &= 30\Omega + 30\Omega + 30\Omega \\ &= 90\Omega\end{aligned}$$

The modified diagram of Figure 8 is shown in Figure 9:

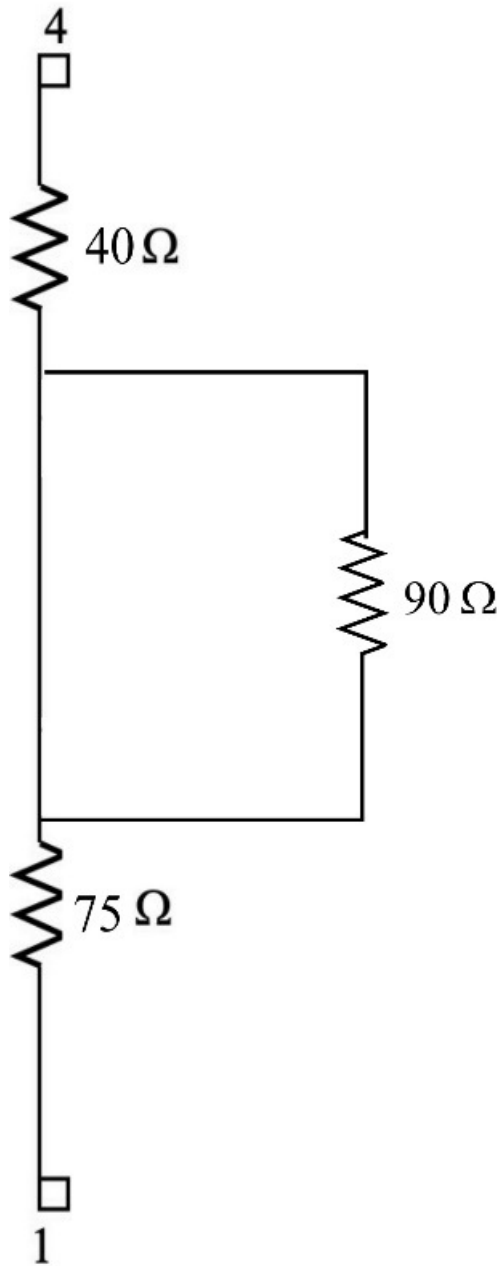


Figure 9

Step 15 of 17

From Figure 9,

The resistance 90Ω is sorted (0Ω) .

Calculate the equivalent resistance is as follows.

$$\begin{aligned}(90\Omega) \square (0\Omega) &= \frac{(90\Omega)(0\Omega)}{(90\Omega) + (0\Omega)} \\ &= 0\Omega\end{aligned}$$

Step 16 of 17

The modified diagram of Figure 9 is shown in Figure 10:



Figure 10

Step 17 of 17

From Figure 10,

Calculate the equivalent resistance between 1 and 4 are as follows.

$$\begin{aligned}\text{Equivalent resistance} &= 75\Omega + 40\Omega \\ &= 115\Omega\end{aligned}$$

Therefore, the equivalent resistance between 1 and 4 is 115Ω.